Hertie School

When Rough Seas Become Deadlier: Migrant Mortality and the 2017 Italy–Libya Memorandum

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Abstract

The Central Mediterranean is the world's deadliest migration route, yet quantitative evidence on how the post-2017 EU migration policy regime changed the risk faced by migrants attempting the journey from North Africa to Europe remains limited. This thesis asks whether the same sea-state conditions are associated with more recorded migrant deaths on the Central Mediterranean after the 2 February 2017 Italy–Libya Memorandum of Understanding (MoU) than before, using a Central Mediterranean route-level panel from January 2014 to May 2023. In the primary negative-binomial model with month–year fixed effects, a one-metre rise in the lagged five-day mean of significant wave height (SWH) corresponds to a roughly 94% decrease in expected recorded deaths before the MoU but more than doubles them after (pre-MoU slope -2.829, post-MoU shift $+3.629$); the shift survives controls for crossing volume, boat composition, and conflicts in Libya and Tunisia. Mechanism checks point to search-and-rescue availability: in weeks with higher recorded SAR activity rough seas predict fewer additional deaths, and the post-MoU gap returns toward zero during the Lamorgese partial rollback. The findings reframe assessment of post-2017 sea-border policy: after the regime change the same sea conditions are associated with substantially more recorded deaths, and SAR availability buffers that association.

1 Introduction

The Central Mediterranean is the deadliest migration route in the world. According to the IOM Missing Migrants Project (International Organization for Migration, 2026), the corridor between Libya, Tunisia, and Algeria and the southern shores of Italy and Malta has recorded 34,900 dead and missing migrants since 2014, with counts current to 16 May 2026. The UNITED for Intercultural Action *List of Refugee Deaths* (UNITED for Intercultural Action, 2026) counts 71,337 documented fatalities attributable to the "Fatal Policies of Fortress Europe" since 1993, and many more are never recorded. No other contemporary migration route has comparable mortality.

Some of these deaths occurred while EU migration policy shifted away from state-led rescue and toward externalised interception and constraints on non-governmental search-and-rescue (SAR). The Memorandum of Understanding (MoU) between Italy and Libya[1], signed the 2 February 2017, marks this shift on the Central Mediterranean Route. The MoU committed Italy to financing and equipping the so-called *Libyan Coast Guard* to intercept migrants at sea and return them to Libya. A sequence of Italian and EU measures then followed, restricting the scope for civilian rescue (Cusumano & Villa, 2021; Moreno-Lax, 2018; Punzo & Scaglione, 2024).

The empirical question this thesis asks is how this policy changed the risk faced by those crossing the Central Mediterranean. The political case for the regime rests in part on the claim that non-governmental organization- (NGO) and state-led rescue acted as a "pull factor", so that cutting rescue while expanding interception would reduce crossings and, mechanically, deaths. Recent quantitative work has weakened this claim on the volume side (Hoffmann Pham & Komiyama, 2024; Rodríguez Sánchez et al., 2023; Tafani & Riccaboni, 2025). What remains open is the *per-crossing risk*: how likely a given crossing is to end in death, conditional on departure (Zambiasi & Albarosa, 2025).

This thesis addresses that gap. The research question is: how does the association between sea state and recorded deaths change across the pre- and post-MoU regimes? The estimand is the slope linking sea state to recorded mortality, and how this slope shifts across the regime change. Figure 1 places the Central Mediterranean within the broader system of Mediterranean and Atlantic routes; closure on one corridor has historically displaced both flow and risk to others (Hoffmann Pham & Komiyama, 2024; Tafani & Riccaboni, 2025; Vives, 2023).

The empirical strategy uses daily variation in significant wave height (SWH) as a plausibly exogenous sea-condition hazard, conditional on month–year fixed effects, tracing how the same conditions map into recorded mortality before and after the MoU. The January 2014–May 2023 panel combines ERA5 sea state, UNITED and IOM mortality

[1]Governo Italiano–Presidenza del Consiglio dei Ministri, governo.it/sites/governo.it/files/Libia.pdf, Memorandum d'intesa sulla cooperazione nel campo dello sviluppo, delcontrasto all'immigrazione illegale, al traffico di esseri umani, al contrabbando e sul rafforzamento della sicurezza delle frontiere tra lo Stato della Libia e la Repubblica Italiana, 2 February 2017.

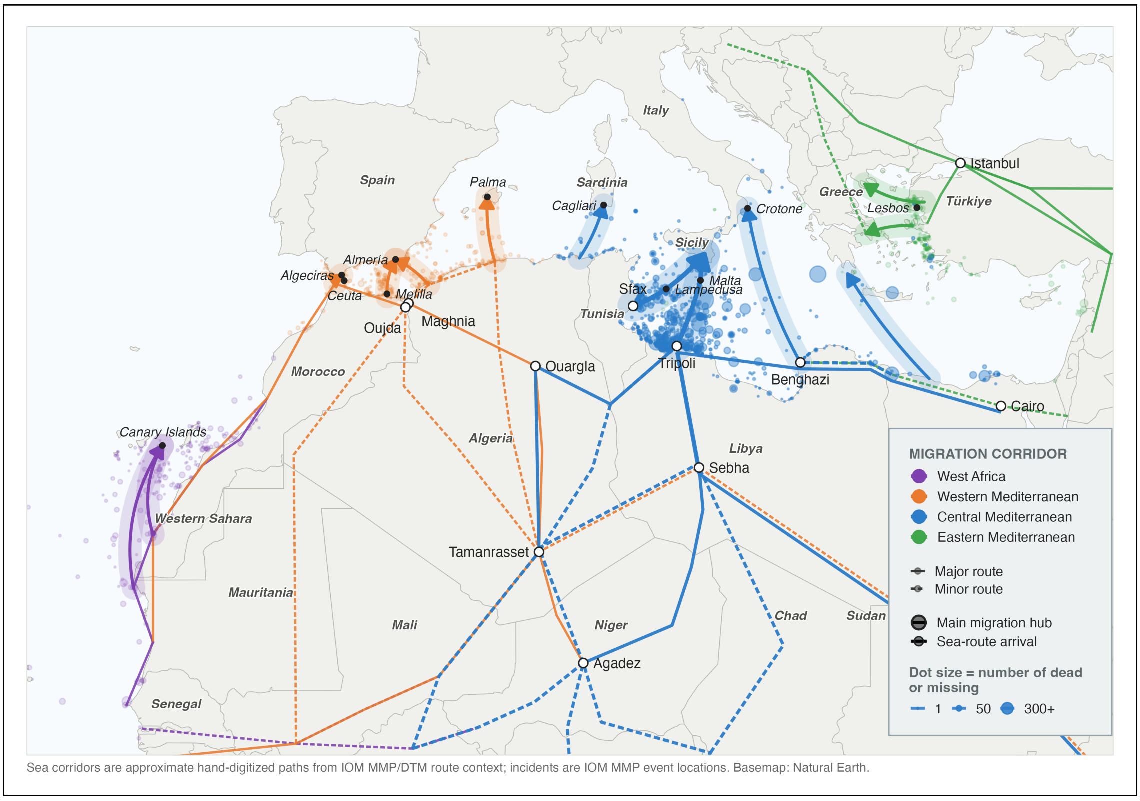

Figure 1: Mediterranean and Atlantic migration routes, dead and missing migrant events, and main route hubs, 2014–2023. Source: own elaboration. Data: approximate hand-digitized sea routes from IOM MMP/DTM route context; IOM Missing Migrants Project event locations; Natural Earth base map.

records, Frontex incident microdata, and Libyan and Tunisian coast-guard pullbacks. After restricting deaths to the corridor polygon and sea-crossing causes, the sample contains 20,368 UNITED deaths and 15,285 IOM dead-and-missing. Because there is no untreated comparison route, the design estimates a within-route change in weather sensitivity rather than a difference-in-differences treatment effect.

The weather–mortality gradient shifts sharply upward around the MoU. In the primary UNITED negative-binomial model, the pre-MoU SWH slope is -2.829 (SE 0.723) and the post-MoU shift is $+3.629$ (SE 0.820): rough seas predicted fewer recorded deaths before the MoU, but more recorded deaths after it. The IOM series reproduces the same pattern, and the shift survives controls for crossing volume, boat composition, and a Libya conflict composite. Mechanism checks point in the same direction: weeks with more recorded Frontex SAR activity show a flatter SWH–mortality slope. After the MoU, rough seas became more lethal in the recorded data, while greater rescue availability appears to have buffered that relationship.

The remainder of the paper reviews the literature, develops the estimand, describes the data and identification strategy, reports the main and mechanism results, and concludes.

2 Literature Review

A substantial literature shows that intensified or externalized border enforcement can raise migrant mortality even when it reduces departures, either by displacing migrants onto more hazardous routes or by making a given crossing more dangerous. The canonical case is the United States border, where tighter controls after 1993 redirected crossings into dangerous desert terrain and increased the death toll (Cornelius, 2001; Solano & Massey, 2022). A similar relationship is observed at sea: EU enforcement operations between 2006 and 2015 are positively associated with fatalities and rerouting toward more dangerous, less-patrolled crossings (Williams & Mountz, 2018). Likewise, post-2018 Western Mediterranean agreements increased mortality by militarizing the border and pushing migrants onto longer routes (Prieto-Flores, 2025; Vives, 2023).

Border policy can affect sea mortality through three channels: the number of crossings (*deterrence*), the boats, routes, and timing chosen by smugglers and migrants (*moral hazard*), and the risk of dying once boats are at sea (*risk-reduction*). The first two channels are partly visible in administrative records and have been studied empirically for the Central Mediterranean (Deiana et al., 2024; Hoffmann Pham & Komiyama, 2024; Rodríguez Sánchez et al., 2023). The third is harder to observe directly, but it is the one on which SAR should matter most.

Crossing volume shows no detectable response to SAR itself, contrary to the "pull factor" hypothesis; coordinated Libyan Coast Guard pushbacks did reduce attempted crossings (Rodríguez Sánchez et al., 2023). Boat composition responded more clearly through a moral-hazard channel: the earlier expansion of SAR encouraged smugglers to use flimsier rafts and depart in worse weather, offsetting part of the intended safety gain (Deiana et al., 2024). Route choice also adjusted: the contraction of rescue deterred Central Mediterranean crossings and diverted part of the flow toward the Western route (Hoffmann Pham & Komiyama, 2024), a cross-route displacement also observed at measurable cost in lives after the EU–Turkey agreement (Tafani & Riccaboni, 2025).

This behavioral evidence matters for identification. Weather is plausibly exogenous, but exposure to weather is not mechanically fixed: rough seas affect departures (Camarena et al., 2020; Deiana et al., 2024), and departure timing is partly organized by smugglers rather than by migrants alone (Camarena et al., 2020). Smugglers also adjust boat technology, crowding, and route strategy when rescue or interception probabilities change (Deiana et al., 2024; Hoffmann Pham & Komiyama, 2024). A post-2017 change in the weather–mortality gradient can therefore reflect the weakening of the SAR buffer, endogenous changes in the crossings exposed to rough seas, or both. The empirical analysis below treats these behavioral channels as central threats to a narrow SAR interpretation rather than as background noise.

Quantitative studies of migrant mortality at sea are few. One remaining gap is how the risk of death during an attempted Central Mediterranean crossing changes across policy periods as SAR availability contracts. Volume, composition, and route diversion describe how many people cross and under what conditions, not the risk itself. Spatial evidence on the MoU locates where deadly incidents became more likely rather than tracing their relationship to sea state over time (Zambiasi & Albarosa, 2025); sea state has been used as exogenous variation on this route, but to explain crossing behavior rather than mortality (Deiana et al., 2024).

This thesis fills that gap by treating the slope linking sea state to recorded deaths as the estimand. The main analysis estimates how the whole post-MoU interception regime changed this slope. It then tests whether one mechanism modified by that regime, SAR availability, moves the slope in the direction predicted by the risk-reduction account.

3 Theoretical Framework and Empirical Strategy

3.1 Externalized border control and the reconfiguration of search-and-rescue

This thesis focuses on the 2017 Italy–Libya Memorandum of Understanding (MoU) because it made a longer trajectory of externalized border control operationally central to rescue and interception on the Central Mediterranean route. Italy–Libya and EU–Libya cooperation on patrols, expulsions, detention, and border-management support had existed for two decades (Bialasiewicz, 2012; Lutterbeck, 2006; Paoletti, 2011), and outsourcing migration management to transit states had been identified as a policy paradigm since the early 2000s (Boswell, 2003). After the 2015–2016 crisis, externalization became more intensive and institutionalized through interstate instruments that delegated interception to third-country authorities while seeking to insulate European states from direct legal responsibility (Frelick et al., 2016; Moreno-Lax & Giuffrr, 2017; Müller & Slominski, 2021; Niemann & Zaun, 2023). Figure 2 traces this sequence from the state-led rescue model of 2013–2014, through the 2017 MoU, to later containment of NGO operations and measures obliging NGO vessels to disembark to far ports, rather than the closest available safe European port.

The 2017 MoU was the Central Mediterranean expression of this wider shift. Signed on 2 February 2017, it committed Italy to supporting Libyan border-control infrastructure, training, and equipment, including for the Libyan Coast Guard (LCG); the Malta Declaration[2] endorsed the same approach at EU level. Its logic echoed the 2016 EU–Turkey Statement[3] and positioned the MoU within a broader EU turn toward externalized and securitarian border governance (Ghezelbash et al., 2018; Wolff, 2024).

This institutional shift also changed the practical organization of search-and-rescue (SAR). The direct state-led rescue model of the Italian Navy's Operation Mare Nostrum (October

[2]European Council, consilium.europa.eu/en/press/press-releases/2017/02/03/malta-declaration/, Malta Declaration by the members of the European Council on the external aspects of migration: addressing the Central Mediterranean route, 3 February 2017.

[3]European Council, consilium.europa.eu/en/press/press-releases/2016/03/18/eu-turkey-statement/, EU-Turkey statement, 18 March 2016.

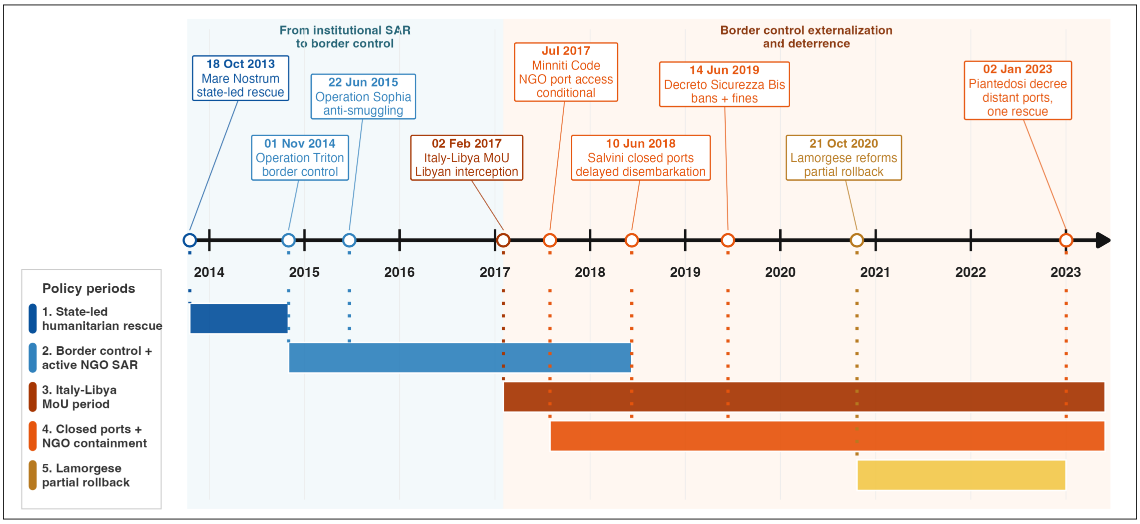

Figure 2: Central Mediterranean rescue, externalization, and NGO-containment policy timeline. Source: own elaboration.

2013–November 2014) had already given way to a more limited model centered on border surveillance and counter-smuggling under Frontex's Operation Triton, launched in late 2014, and EUNAVFOR MED Sophia, launched in 2015 (Cusumano, 2019a). NGO vessels increasingly filled the resulting rescue gap, especially in international waters near the Libyan coast (Cusumano & Villa, 2021).

The year 2017 was also pivotal for the containment of NGO SAR operations. That year, the Italian government issued a Code of Conduct[4] for NGOs engaged in the rescue of migrants at sea. This framework was later hardened by the 2019 Salvini security decrees, notably Decree-Law No. 53[5] ("Decreto Sicurezza Bis"), which authorized restrictions on the entry, transit, and stay of rescue vessels in Italian territorial waters and introduced heavy fines for non-compliant ships (Cusumano, 2019b; Cusumano & Villa, 2021). Although the Lamorgese Decree-Law No. 130 of 21 October 2020[6] partially rolled back this framework by reducing the severity of sanctions and restoring some protection and reception guarantees, it did not abandon the underlying approach to migration governance. This logic continued under later governments through administrative measures such as the assignment of distant ports of disembarkation, which increased the time NGO vessels spent away from the rescue zone and further reduced rescue availability (Punzo & Scaglione, 2024).

[4] Governo Italiano–Ministero dell'Interno, interno.gov.it/it/amministrazione-trasparente/disposizioni-generali/atti-generali/atti-amministrativi-generali/decreti-direttive-e-altri-documenti/codice-condotta-ong-impegnate-nel-salvataggio-dei-migranti-mare, Codice di condotta per le ONG impegnate nel salvataggio dei migranti in mare, 7 August 2017.

[5] Governo Italiano–Presidenza del Consiglio dei Ministri, normattiva.it/uri-res/N2Ls?urn:nir:stato:decreto.legge:2019;53, Disposizioni urgenti in materia di ordine e sicurezza pubblica, 14 June 2019.

[6] Governo Italiano–Presidenza del Consiglio dei Ministri, normattiva.it/uri-res/N2Ls?urn:nir:stato:decreto.legge:2020-10-21;130, Disposizioni urgenti in materia di immigrazione, protezione internazionale e complementare, modifiche agli articoli 131-bis, 391-bis, 391-ter e 588 del codice penale, nonchè misure in materia di divieto di accesso agli esercizi pubblici ed ai locali di pubblico trattenimento, di contrasto all'utilizzo distorto del web e di disciplina del Garante nazionale dei diritti delle persone private della libertà personale, 14 June 2019.

European state-led rescue availability receded in parallel. Operation Sophia's naval assets were suspended in March 2019, and from 2019 the Italian Coast Guard increasingly reclassified search-and-rescue situations as law-enforcement operations (Düvell & Denaro, 2024). More recently, the EU New Pact on Migration and Asylum[7] has placed externalization and return as central logics of irregular migration governance (Wolff, 2024).

Moreno-Lax (2018) describes the resulting EU paradigm as one of "rescue-through-interdiction / rescue-without-protection": humanitarian language is used to legitimize interception, while rescue is reduced to keeping migrants alive long enough for them to be intercepted and taken back to Libya, rather than brought to a place where they can claim protection. The MoU operationalized this paradigm by financing and equipping the actor that performs the interception, thereby displacing the practical enforcement of non-refoulement obligations onto a third party.

3.2 Weather, SAR, and per-crossing risk

SAR does not eliminate the hazards of the crossing — precarious boats, overcrowding, and prolonged exposure at sea — but it can prevent those hazards from escalating into shipwrecks and deaths (Cusumano & Gombeer, 2020; Cusumano & Riddervold, 2023; Kosmas et al., 2022). As European state-led SAR assets withdrew from international waters off Libya, and as NGO rescue operations became increasingly constrained, the risk-reduction effect of SAR is expected to have weakened.

One of the most evident hazards migrants face at sea is adverse weather. Qualitative accounts identify stormy weather and rough seas as key causes of death, alongside overloading, fuel shortages, and engine failure (Idemudia & Boehnke, 2020). Existing evidence suggests that bad weather reduces departures from North African coasts towards Europe (Camarena et al., 2020; Deiana et al., 2024), while crossings that do occur in rough weather likely carry higher fatality risk.

The theoretical mechanism examined in this thesis is whether rescue availability moderates the effect of rough sea conditions on the risk of death. Sea state affects recorded deaths through two distinct channels. The first channel operates through departures: when significant wave height (SWH) is high, fewer migrants depart from the North African coasts, which mechanically reduces the recorded number of deaths (*deterrence* – or *volume channel*). The second channel operates through per-crossing lethality: conditional on having departed, rough seas increase the probability of distress, capsizing, and death (the *per-crossing risk channel*).

The MoU moderates this second channel primarily through rescue availability—the extent to which a vessel in distress can be located, reached, rescued, and disembarked in a European place of safety. It may also operate through the quality of boats used by smugglers, levels

[7] European Commission, home-affairs.ec.europa.eu/policies/migration-and-asylum/pact-migration-and-asylum_en, Pact on Migration and Asylum, 20 May 2024.

of overcrowding, route choice, and departure timing. When rescue assets are nearby, able to respond quickly, and permitted to disembark survivors at the nearest safe port, they partly buffer the lethality of the journey; when that buffer weakens, the same physical hazard is more likely to result in a recorded death.

SWH is plausibly exogenous to the migration-control regime. This is why weather provides a useful bridge between the theory and the empirical strategy. Within a policy regime, daily variation in lagged SWH traces how sensitive recorded mortality is to sea hazard under the prevailing rescue configuration. Across regimes, the change in that sensitivity, a shift in the slope linking sea state to recorded mortality, is the analytical target.

3.3 Theoretical estimand

Following Lundberg et al. (2021), defining the theoretical estimand—the target quantity implied by the research question, before any choice of estimator—clarifies what the empirical strategy can and cannot deliver.

Let D_t denote recorded deaths on day t, $s_t \equiv SWH_{t-1:t-5}$ the lagged five-day mean of SWH, and R_t rescue availability. The expected recorded count factors into three terms, each a function of sea state and rescue availability:

$$\mathbb{E}[D_t \mid s_t, R_t] \;=\; q(s_t, R_t)\, A(s_t, R_t)\, m(s_t, R_t). \tag{1}$$

$q(s_t, R_t)$ is *recording*: the probability that a death is captured in the mortality data. $A(s_t, R_t)$ is *exposure*: the expected number of crossing attempts that day, counted per person, so that one migrant's departure is one attempt. $m(s_t, R_t)$ is *per-attempt lethality*: the latent probability that a crossing attempt ends in death, conditional on departure. Exposure and lethality are the two behavioral channels of the preceding subsection; recording is a distinct measurement channel, present because the outcome is *recorded* rather than actual deaths. The product $A\,m$ is the expected number of actual deaths, and $q\,A\,m$ the expected number of recorded deaths.[8]

The theoretical estimand is the cross-partial $\partial^2 \log m/\partial s\, \partial R$, which measures how rescue availability moderates the slope linking rough seas to lethality. The risk-reduction hypothesis predicts that this quantity is negative: greater rescue availability flattens the SWH–lethality slope. Because rescue availability contracts after the MoU, a negative cross-partial implies that the lethality slope steepens across the regime change.

This estimand is not directly observed, because q, A, and m are not separately identified from recorded deaths alone. The observable object is the reduced-form slope of recorded deaths on sea state, the term-by-term logarithmic derivative of Equation 1:

[8] Equation 1 is an identity, not a probabilistic factorization: $A(s_t, R_t)$ is expected attempts, and m and q are the conditional ratios of expected actual deaths to expected attempts and of expected recorded deaths to expected actual deaths.

$$\frac{\partial \log \mathbb{E}[D_t \mid s_t, R_t]}{\partial s_t} \;=\; \underbrace{\frac{\partial \log q}{\partial s_t}}_{\text{recording}} + \underbrace{\frac{\partial \log A}{\partial s_t}}_{\text{exposure}} + \underbrace{\frac{\partial \log m}{\partial s_t}}_{\text{lethality}}. \tag{2}$$

Differentiating Equation 2 with respect to R shows that the *shift* in this slope across rescue regimes is itself a sum of three cross-derivatives, only one of which (the lethality term $\partial^2 \log m/\partial s\,\partial R$) is the theoretical estimand.

Recovering the theoretical estimand from the observed shift therefore requires assumptions on the recording and exposure cross-derivatives. If both are zero, the observed shift identifies the lethality shift exactly; if they move opposite to lethality, it is a lower bound. The moral-hazard response documented by Deiana et al. (2024) implies such an opposite-signed exposure shift: the post-MoU contraction of rescue makes departures more sensitive to rough weather. Recording plausibly moves the same way, because rough-weather shipwrecks are least likely to leave survivors or a recoverable wreck, so reduced rescue should degrade recording more in rough weather than in calm. Overstatement would require recording or exposure sensitivity to steepen in the same direction as lethality after 2017. The constructed-exposure control introduced below, the boat-composition checks in Appendix F, and two independently collected mortality sources discipline that case.

3.4 From theoretical to empirical estimand

The reduced-form slope shift derived above is the observable counterpart that can be estimated with the available data. Operationally, the empirical estimand is the change in the SWH-recorded-death slope around the post-MoU regime:

$$\beta_3 \;=\; \beta_{\text{post}} \;-\; \beta_{\text{pre}}, \tag{3}$$

where β_{pre} and β_{post} are the slopes of log expected deaths on the lagged five-day mean of SWH before and after 2 February 2017. For example, negative pre-period slopes are consistent with rough seas limiting departures when rescue availability is greater.

The β_3 term is a pre/post difference in the conditional SWH gradient. It is algebraically analogous to the interaction term in a continuous-regressor DiD, but without an untreated comparison unit it is not a standard DiD treatment effect. It identifies a within-route change in the conditional SWH-recorded-mortality slope: a policy-regime moderation parameter for the whole post-MoU regime.

A causal interpretation of this regime-level estimate would require a credible counterfactual for how the same route's SWH–mortality gradient would have evolved absent the regime; the design observes only one route before and after the policy shift. The estimate is not a count of deaths prevented by SAR, nor a direct estimate of the causal effect of SAR availability on mortality, which would require otherwise similar crossings that differ exogenously in rescue availability. The post-MoU regime bundled changes in rescue availability, interceptions, NGO operations, smuggler incentives, and recording conditions; this bundling is part of the regime being estimated, but it limits attribution to any single component.

Moreover, rescue availability R_t is latent: no public dataset measures the stock of rescue assets at sea. The empirical strategy first estimates the regime-level slope shift, then uses weekly Frontex SAR measures to test whether a SAR-related mechanism is consistent with it. These proxies capture recorded rescue activity, not random assignment of rescue capacity.

3.5 Research question and hypotheses

This framing yields a single empirical question:

> How does the marginal association between sea state and recorded deaths change across the pre- and post-MoU policy regimes?

The estimand for this question is the post-MoU slope shift β_3. The main thesis statement is that the post-MoU interception regime should make the SWH-mortality slope less negative or more positive if it weakened the buffer between rough seas and death. The prediction tested below is therefore $\beta_3 > 0$. A second, mechanism-level prediction is that higher observed SAR activity should flatten the SWH-mortality slope.

The literature documents several behavioral responses to the post-2017 reconfiguration—shifts in crossing volume, boat composition, departure timing, and route choice. The channel this thesis examines, the *per-crossing risk* of dying given a crossing in given sea conditions, is distinct from these but cannot be separated from them without assumptions. The prediction $\beta_3 > 0$ corresponds to a movement in this channel.

If β_3 is absorbed once exposure is controlled, the change is consistent with the volume and diversion channels alone. If it is absorbed once boat composition is controlled, the change is consistent with moral hazard alone. The prediction is that β_3 persists under both; if it does, the per-crossing-risk channel has moved in addition to whatever else has changed.

4 Methods and Data

4.1 Setting and sample

The analysis uses a daily panel for the Central Mediterranean Route, with weather, mortality, and crossing variables measured over the same area. Daily resolution is necessary because sea state changes quickly and can affect both departures and survival at sea. The design follows Deiana et al. (2024), which uses significant wave height (SWH) as an exogenous shock to crossing conditions on this route; monthly aggregation would absorb much of the identifying weather variation.

Figure 3 defines the spatial frame. The analysis polygon is both the area over which daily

SWH is averaged and the filter applied to mortality records. It retains 83.3% of UNITED-recorded CMR deaths and 88.7% of IOM dead-and-missing over January 2014–May 2023. The figure also places the area of analysis in relation to the Italian, Maltese, Libyan, and Tunisian search-and-rescue responsibility zones.

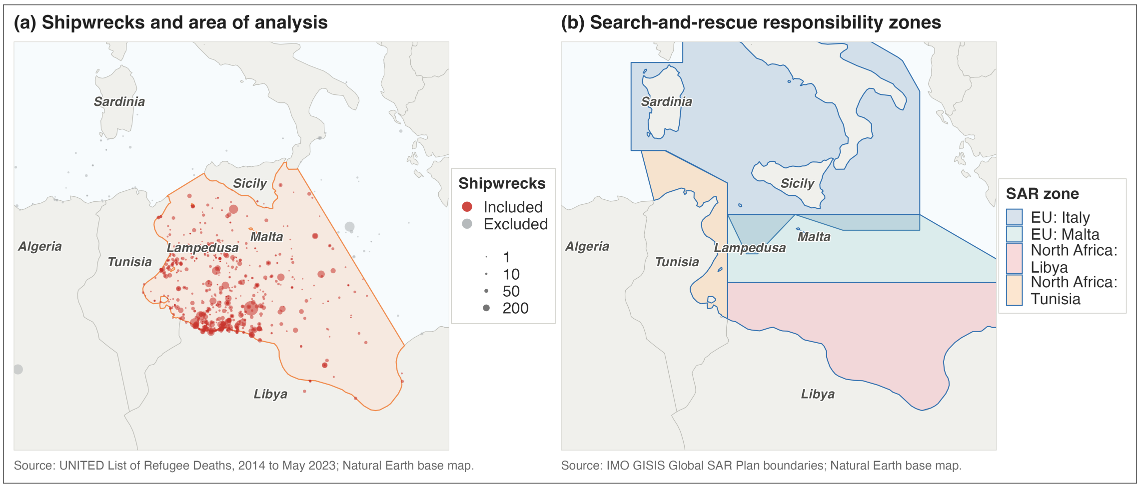

Figure 3: Area of analysis, UNITED deadly incidents, and search-and-rescue responsibility zones. Source: own elaboration. Data: UNITED List of Refugee Deaths, January 2014 to May 2023; IMO GISIS Global SAR Plan boundaries; Natural Earth base map.

Using one corridor-level SWH series is therefore a spatial aggregation choice. Appendix B checks this choice directly by comparing the daily SWH series of all ERA5 ocean cells inside the polygon and by re-estimating the primary count model with SWH weighted toward cells containing more recorded incidents. The average cell has a correlation of 0.863 with the polygon mean, and the primary UNITED slope-shift estimate is nearly unchanged when incident-count weights are used.

The series runs from 1 January 2014 to 31 May 2023, the last complete month with Frontex coverage used to disaggregate Libyan and Tunisian coast-guard pullbacks to daily frequency. The regression sample begins on 15 January 2014, once the lagged crossing window has filled.

The policy-regime indicator is $Post_t = \mathbf{1}\{t \geq 2 \text{ February } 2017\}$, the date the Italy–Libya MoU was signed. The cutoff defines the post-MoU interception regime for the reduced-form comparison, not a sharp causal switch. Period and rolling-window specifications later relax this binary split.

4.2 Data sources

The panel combines ERA5 sea state, UNITED and IOM mortality records, Frontex's Joint Operations Reporting Application (JORA), and Libyan and Tunisian Coast Guard (LCG/TCG) pullbacks from IOM's Global Migration Data Analysis Centre (GMDAC). ACLED conflict events and UNHCR arrivals enter only in robustness checks.

The data used in the models are summarized in Table 1.

Table 1: Summary of data used in the analysis.

Sample window		
Date range		2014-01-01 to 2023-05-31
Calendar days		3,438
Regression sample (lagged windows filled)		3,424
Volume-controlled sample (days with $C_t > 0$)		2,036
Mortality outcomes (analysis sample)		
	UNITED	*IOM*
Deaths (total)	20,368	15,285
Death-days	503	616
Maximum daily count	1,101	1,022
Cross-source correlation, daily		0.71
Cross-source correlation, monthly		0.91
Lower-bound crossing-attempt proxy C_t		
	Count	*Share*
Persons in Frontex-recorded crossing events	819,576	75.7%
Persons pulled back by Libyan/Tunisian coast guards	242,934	22.4%
Deaths recorded by UNITED	20,368	1.9%
Total lower-bound attempts C_t	**1,082,878**	**100.0%**

Notes: Mortality data are filtered by geographic polygonal area and by cause of death (drowning, mixed, or unknown); IOM figures exclude split incidents.

Sea state. Daily SWH is the ERA5 spatial mean (Hersbach et al., 2020) over sea cells in the analysis polygon. The primary regressor is the lagged five-day mean $SWH_{t-1:t-5}$. A five-day window covers the sea state relevant to a day-t recorded death: pre-departure conditions that shape smuggler launch decisions, in-voyage conditions on a crossing that can last from a day to several days, and the delay between an incident and its appearance in either registry. Daily SWH is also strongly autocorrelated (lag-1 $\rho \approx 0.73$), so the five-day mean summarizes persistent maritime hazard rather than smoothing out independent shocks. One-, three-, and seven-day lagged means are used for window-sensitivity checks in Appendix C; all return positive and significant slope shifts on UNITED (Table A2), so the choice of window does not drive the result. All exclude day-t weather.

Mortality. Recorded deaths come from the UNITED *List of Refugee Deaths* and the IOM Missing Migrants Project (Poole et al., 2020; Rodríguez Sánchez et al., 2023). UNITED records are restricted to the analysis polygon, the five Central Mediterranean incident countries plus open-sea records coded as Mediterranean, and deaths by drowning or other/unknown causes. IOM records are restricted to the Central Mediterranean route, the same five countries, incident-type records inside the polygon, and drowning or mixed/unknown causes; split incidents are excluded. UNITED is the primary outcome and IOM a robustness check.

Frontex incident microdata. Frontex JORA provides daily records of interceptions at sea and on European coasts. The 2014–2023 Central Mediterranean extract, released under PAD-194, covers Triton and Themis. The CMR sample retains departures from Libya, Tunisia, or Algeria and records SAR status, Frontex involvement, and the detecting or intercepting actor, allowing NGO-led rescues to be distinguished from interceptions by national assets.

Coast-guard pullbacks. Libyan and Tunisian Coast Guard pullbacks come from IOM-GMDAC monthly Mediterranean migration figures. Monthly totals are allocated to days by proportional Denton disaggregation (Denton, 1971), using Frontex-recorded daily departures from Libya or Tunisia as high-frequency indicators. The result is an approximate volume control, not a treatment variable.

ACLED conflict. A one-week-lagged Libya conflict composite from ACLED enters only the push-factor robustness check.

Figure 4 summarizes monthly crossing outcomes and interception composition over 2014–2023. The post-2017 compositional shift motivates keeping Frontex SAR, Frontex non-SAR, and LCG/TCG categories separate in the panel and combining them only in the lower-bound crossing-attempt series C_t defined in Equation 6.

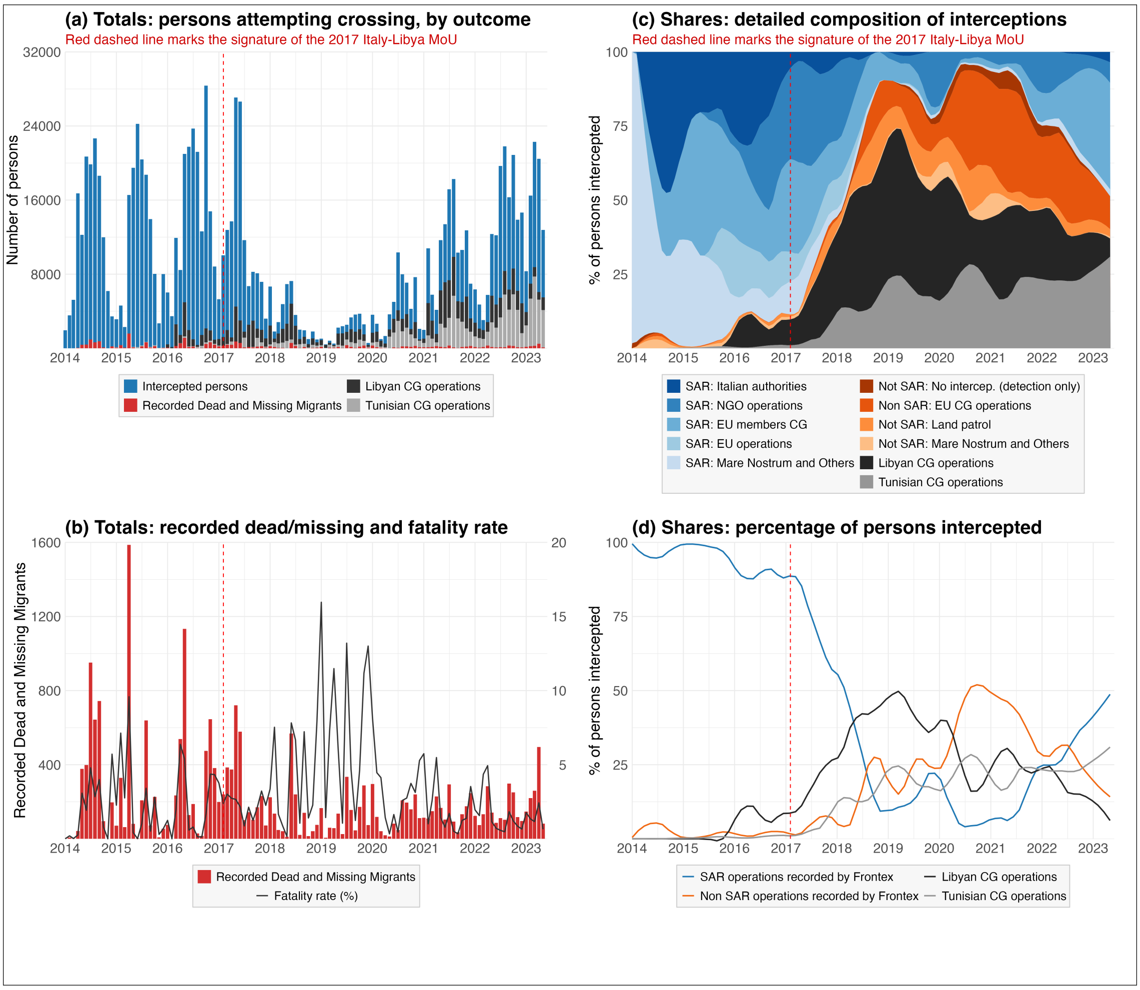

Figure 4: Observed crossing outcomes and interception composition, 2014–2023. Source: own elaboration.

4.3 Outcome and estimand

The outcome is the daily count of recorded deaths in source j, D_t^j, with $j \in \{\text{UNITED}, \text{IOM}\}$. The empirical estimand is the post-MoU slope shift β_3:

$$\beta_3 = \frac{\partial \log \mathbb{E}[D_t^j \mid SWH_{t-1:t-5}, Post_t = 1]}{\partial SWH_{t-1:t-5}} - \frac{\partial \log \mathbb{E}[D_t^j \mid SWH_{t-1:t-5}, Post_t = 0]}{\partial SWH_{t-1:t-5}}. \quad (4)$$

Section 3 predicts $\beta_3 > 0$ if the post-MoU regime weakened the buffer between rough seas and death. Because β_3 identifies a regime-level moderation parameter rather than the effect of any single component, Section 4.5 returns to the identification threats this raises and Section 4.6 to the robustness checks that discipline them.

4.4 Empirical specification

The primary specification estimates daily deaths as

$$\mathbb{E}[D_t^j] = \exp(\alpha_{m(t)} + \beta_1 SWH_{t-1:t-5} + \beta_3 SWH_{t-1:t-5} \times Post_t). \quad (5)$$

The month-year fixed effect $\alpha_{m(t)}$ absorbs seasonality and common monthly shocks. Identification of β_3 relies on within-month variation in lagged sea state and on the pre/post difference in how that variation maps into deaths. The specification does not estimate a separate post-MoU level shift. Results report the pre-period slope β_1, the shift β_3, and the implied post-period slope $\beta_1 + \beta_3$ with a delta-method standard error.

This design is therefore closer to a hazard-sensitivity test than to a before–after comparison of death levels: the interaction term asks whether day-to-day weather sensitivity differs after the post-MoU regime, the dimension on which the SAR mechanism should operate if rescue buffers rough-weather lethality.

The model is estimated by negative-binomial maximum likelihood and Poisson quasi-maximum likelihood. Negative binomial accommodates sparse, overdispersed counts; Poisson QMLE estimates the conditional mean without requiring mean-variance equality. The same specification is estimated on UNITED and IOM, with source-specific all-zero fixed-effect cells dropped by the estimator. Inference uses Newey–West standard errors with a 14-day bandwidth; month-year clustered errors are reported alongside.

Because crossing volume varies over time and may affect recorded deaths independently of sea state, a volume-controlled variant is also estimated. The crossing-volume control is a lower-bound daily count,

$$
\begin{aligned}
C_t = {} & \text{people recorded by Frontex in crossing events} \\
& + \text{people pulled back by Libyan or Tunisian coast guards} \\
& + \text{deaths recorded by UNITED.}
\end{aligned} \tag{6}
$$

where each component is counted on day t. This follows the Central Mediterranean convention of summing arrivals, interceptions, returns, and deaths to approximate attempted crossings (Deiana et al., 2024; Rodríguez Sánchez et al., 2023). The control enters in logs, with its coefficient estimated from the data:

$$
\mathbb{E}[D_t^j] = \exp(\alpha_{m(t)} + \beta_1 SWH_{t-1:t-5} + \beta_3 SWH_{t-1:t-5} \times Post_t + \gamma \log C_t), \tag{7}
$$

estimated on days with $C_t > 0$. This specification asks whether the SWH slope shift remains after accounting for observed crossing volume.

4.5 Identification and threats

The identifying argument has two layers. First, conditional on month-year fixed effects, day-to-day variation in lagged SWH is as-good-as random with respect to other determinants of recorded mortality. Second, the policy variable must be interpreted as the wider interception regime initiated by the MoU—expanded Libyan interceptions, NGO restrictions, and later Italian legal and administrative measures—not the legal memorandum alone. The main counterfactual problem is that the design does not observe how the same route's weather–mortality gradient would have evolved absent that regime.

The sharpest omitted-variable threat is that the composition of crossings exposed to a given SWH may change endogenously across regimes. Let B_t collect unobserved behavioral variables such as departure timing, route choice, boat quality, overcrowding, and choices around when to make a distress call. If B_t is correlated with $SWH_{t-1:t-5} \times Post_t$ and also affects recorded deaths, β_3 mixes the SAR risk-reduction channel with behavioral adaptation. For the broad policy-regime estimand, such adaptation is part of the total regime response. For a narrower claim about SAR availability alone, it is omitted-variable bias.

The behavioral literature makes this threat substantive. Camarena et al. (2020) show that rougher seas reduce Mediterranean arrivals and that smugglers help decide when boats leave. Deiana et al. (2024) show that SAR changed the weather sensitivity of departures and the use of unsafe boats. Hoffmann Pham & Komiyama (2024) show that the post-2017 interception regime affected crossings, diversion, and smuggler choices over boat size. The post-MoU interaction is therefore a reduced-form moderation parameter, interpreted through diagnostics on volume, composition, diversion, conflict, and reporting. These diagnostics can show whether observed proxies absorb the shift; they cannot prove that all behavioral margins are observed.

Incomplete recording is the second major threat. The outcome includes only deaths that enter UNITED or IOM, and observed exposure captures only attempts detected by Frontex or intercepted by Libyan or Tunisian authorities. In the notation of Equation 1, $q(s_t, R_t)$ is below one and $A(s_t, R_t)$ is only partially observed through C_t. The post-MoU shift in β_3 is contaminated if recording changes differentially by sea state across regimes. The likely direction is attenuating: rough-weather shipwrecks are least likely to leave survivors or recoverable wrecks, so reduced rescue after 2017 should degrade recording more in rough weather than in calm. Estimating UNITED and IOM in parallel helps test whether the shift is driven by one registry's recording channel.

4.6 Mechanism and robustness specifications

The mechanism specification asks whether one component modified by the post-MoU regime, SAR availability, moves the weather–mortality gradient in the predicted direction. It replaces the binary policy interaction with continuous SAR-capacity moderators:

$$\mathbb{E}[D_t^j] = \exp(\alpha_{f(t)} + \theta_1 SWH_{t-1:t-5} + \theta_2 R_{t-1:t-7} + \theta_3 SWH_{t-1:t-5} \times R_{t-1:t-7}), \qquad (8)$$

where $R_{t-1:t-7}$ is a standardized weekly lagged SAR measure. Two versions are used: the previous-week share of Frontex incidents classified as SAR, whose monthly evolution is visible in the SAR-share line of panel (d) in Figure 4, and the log of one plus persons recorded in Frontex SAR events over the same week. Both proxies are endogenous to crossings, distress and operational choices, so they are mechanism checks rather than alternative identification strategies for the regime effect.

Robustness checks target the main threats: spatial aggregation of sea state (Appendix B), past and future SWH windows (Appendix C), the UNITED–IOM cross-source comparison (Appendix D), fixed-effect and inference variants (Appendix E), boat-composition controls (Appendix F), Frontex SAR versus non-SAR event placebos (Appendix G), and ACLED push-factor decompositions (Appendix H). Section 5 also reports period and rolling-window specifications. These checks do not make the design experimental; they discipline the interpretation of β_3 against specific alternatives.

5 Analysis and Results

5.1 Reduced-form slope shift around the MoU

Table 2 reports the main count estimates of Equation 5 on the daily route series. The coefficients are semi-elasticities: they describe the change in log expected recorded deaths associated with a one-metre increase in the lagged five-day mean of SWH, conditional on month–year fixed effects. In multiplicative form, a one-metre rise in lagged SWH scales expected daily deaths by $\exp(\beta)$:

$$\frac{\mathbb{E}[D_t \mid SWH_{t-1:t-5} = s+1]}{\mathbb{E}[D_t \mid SWH_{t-1:t-5} = s]} = \exp(\beta), \qquad \Delta\% = 100 \times [\exp(\beta) - 1]\%.$$

In the UNITED NegBin column, the pre-MoU slope of -2.829 gives $\exp(-2.829) \approx 0.059$—a roughly 94% decrease in expected recorded deaths per additional metre of lagged SWH—while the implied post-MoU slope of $+0.800$ gives $\exp(0.800) \approx 2.23$, an expected count more than doubled. UNITED is the primary outcome; IOM is estimated with the same specification and used as an independent comparison.

For UNITED, the pre-MoU SWH slope is negative in both model families and distinguishable from zero. In the recorded daily count, the departure-reduction channel dominates the conditional-lethality channel: rougher seas suppress departures faster than they raise lethality, and recorded deaths fall on net.

The interaction with the post-MoU indicator is positive in both model families. The observed weather–mortality slope thus changes sign across the regime cutoff: rough seas predict fewer recorded deaths before the MoU, but no longer do so after it. This is the empirical estimand β_3 defined in Equation 3: the reduced-form change associated with the whole post-MoU regime. By itself, it does not isolate SAR, because recorded deaths also reflect exposure, recording, and endogenous behavior. Its sign is nevertheless the sign predicted by the theoretical framework if the regime weakened the rescue buffer between rough seas and death.

This interpretation requires caution. A positive β_3 could also arise if, after 2017, smugglers and migrants changed departure timing, routes, boat quality, crowding, or rescue-seeking behavior in ways that made rough-weather crossings differently selected. The checks below ask whether observed exposure, boat composition, conflict, and reporting differences explain the shift; they do not eliminate unobserved behavioral channels.

The IOM comparison shows the same sign pattern: pre-MoU slopes are negative and post-MoU shifts positive in all four source–model pairs. The implied post-MoU slope ($\beta_1 + \beta_3$) lies close to zero, consistent with a gradient that loses its strongly negative pre-MoU level rather than reversing into a specific positive value.

The two estimators answer the same mean-model question under different distributional assumptions: the negative binomial allows overdispersion in the sparse daily-death series, while the Poisson QMLE relies only on the conditional mean. Both deliver the same conclusion: the SWH–mortality slope shifts upward around the post-MoU interception regime.

UNITED is the more stable daily series. The sources correlate tightly at monthly frequency but less so day by day (Appendix D); IOM records fewer total deaths over more nonzero days and produces one significant future-weather placebo in Appendix C, where UNITED placebos are uniformly null. This isolated leakage is consistent with strong SWH

Table 2: Primary count estimates of the SWH–mortality slope shift around the 2 February 2017 MoU.

	UNITED		IOM (comparison)	
	NegBin	Poisson	NegBin	Poisson
β_1: $\text{SWH}_{t-1:t-5}$	-2.829^{***}	-1.383^{**}	-2.108^{**}	-2.028^{*}
	(0.723)	(0.484)	(0.694)	(0.790)
β_3: $\text{SWH}_{t-1:t-5}$ × Post-MoU	$+3.629^{***}$	$+1.632^{**}$	$+2.478^{**}$	$+2.053^{*}$
	(0.820)	(0.555)	(0.785)	(0.840)
$\beta_1 + \beta_3$: implied post-MoU slope	$+0.800^{*}$	+0.248	+0.370	+0.025
	(0.388)	(0.275)	(0.368)	(0.292)
Month–year fixed effects	Yes	Yes	Yes	Yes
Newey–West SEs (lag 14)	Yes	Yes	Yes	Yes
Observations (days)	3,197	3,197	3,286	3,286
Pseudo R^2	0.021	0.195	0.038	0.240

Stars denote two-sided p-values: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
Newey–West standard errors in parentheses; delta-method SEs for $\beta_1 + \beta_3$.

autocorrelation (lag-1 $\rho \approx 0.73$): the next day's sea state shares signal with the lagged five-day window. The looser daily alignment also reflects IOM's case-based recording, which aggregates heterogeneous sources and indexes entries by reporting date rather than incident date (Poole et al., 2020). IOM therefore plays a corroborating role in what follows.

5.2 Exposure and composition controls

A first alternative interpretation is that β_3 mainly reflects changes in the number of crossing attempts. If the post-MoU shift were only an exposure result, conditioning on the lower-bound crossing-volume measure C_t should absorb much of it. Table 3 shows that it does not. The volume-controlled Poisson estimates retain a negative pre-MoU slope and a positive post-MoU shift in both mortality sources.

The coefficient on crossing volume is positive in both series and well below one. The Wald test, which measures how far an estimated coefficient sits from a hypothesized value in units of its standard error, rejects $\gamma = 1$ in both series, the value that would obtain if crossings entered as a pure exposure offset. C_t is therefore an observed-exposure control, not a full offset for the population at risk. The weather-regime shift remains after adjusting for it, so observed exposure does not explain the result.

A second alternative is boat composition. If the post-MoU regime changed the type of boats used on the route, and if those boat types respond differently to rough seas, composition could mimic a SAR-related shift in the SWH–mortality gradient. The observed change points in the opposite direction: the Frontex-observed inflatable share falls substantially after the MoU. In light of the moral-hazard channel in Deiana et al. (2024), this change should, if anything, attenuate the weather–death relationship as smugglers move away from rough-weather inflatable departures.

The boat-composition checks do not absorb the slope shift. On the boat-observable sample, adding inflatable and wooden shares as additive controls leaves the shift essentially unchanged. Adding a SWH × inflatable-share interaction also leaves the post-MoU shift

Table 3: Volume-controlled Poisson model with crossing-volume control.

	UNITED	IOM (comparison)
β_1: $\mathrm{SWH}_{t-1:t-5}$	-1.390^{**}	-1.911^{*}
	(0.527)	(0.867)
β_3: $\mathrm{SWH}_{t-1:t-5} \times$ Post-MoU	$+1.986^{**}$	$+2.342^{*}$
	(0.608)	(0.933)
γ: $\log C_t$	$+0.377^{***}$	$+0.200^{**}$
	(0.065)	(0.072)
$\beta_1 + \beta_3$: implied post-MoU slope	$+0.596^{*}$	$+0.431$
	(0.293)	(0.326)
Wald: $H_0 : \gamma = 1$, z	-9.55	-11.10
p-value	< 0.001	< 0.001
Month–year fixed effects	Yes	Yes
Newey–West SEs (lag 14)	Yes	Yes
Observations (days, $C_t > 0$)	1,933	1,949

Stars denote two-sided p-values: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
Newey–West standard errors in parentheses; delta-method SEs for $\beta_1 + \beta_3$.

intact, while the interaction itself is not separately identified. Appendix Table A4 reports the full sequence. The post-MoU change in observed boat composition therefore does not account for the estimated slope shift, although unobserved boat quality, overcrowding, route choice, and departure timing could still contribute to it.

Two further checks address residual push-factor variation and the Mare Nostrum baseline. Adding the log of one plus daily ACLED conflict events in Libya and Tunisia, alongside the lag-14 crossing control, leaves the shift essentially unchanged. The ACLED level coefficients are not separately interpretable in this specification: the conflict series is broadcast at weekly resolution and month–year fixed effects absorb most of the relevant variation.

The pre-MoU baseline could in principle be driven by the Mare Nostrum sub-period, when Italian state-led search-and-rescue was at its most active. Restricting the sample to the later Triton/Sophia phase yields essentially the same shift. The slope reversal does not require the Mare Nostrum window.

5.3 Mechanism: continuous SAR proxies

The main estimate treats the post-MoU period as a regime-level change. The mechanism specification asks whether one component modified by that regime, SAR availability, is consistent with the risk-reduction interpretation. If SAR is part of the channel, higher SAR availability should flatten the relationship between rough seas and recorded deaths. Equation 8 tests this implication by replacing the post-MoU indicator with the two standardized, weekly-lagged SAR proxies defined in Section 4.6: the share of Frontex incidents classified as SAR, and the log of one plus persons in Frontex SAR events. Interaction coefficients are per one standard-deviation change in the proxy.

Both moderators yield negative SWH $\times$ SAR-proxy interactions across model families and mortality sources, with most coefficients significant at conventional levels (Table 4). The pattern appears in the primary UNITED series and in the IOM comparison series. This is the direction predicted by the risk-reduction mechanism: in weeks with more recorded SAR activity, the SWH–mortality slope is flatter.

The two SAR proxies have different weaknesses, so their agreement is useful. The SAR share can fall mechanically when non-SAR events increase, while the SAR-persons measure is less exposed to that denominator problem. Same-signed interactions across both measures suggest the pattern is not only a Frontex-denominator artifact. These are mechanism checks rather than causal estimates of SAR deployment, because recorded SAR activity is endogenous to crossings, distress, detection, and operational choices. They support SAR capacity as a plausible channel; they do not estimate deaths prevented by rescue.

Table 4: SWH × SAR-proxy moderator interaction. Coefficients are per one standard deviation of the standardized weekly SAR proxy.

	UNITED		IOM (comparison)	
	NegBin	Poisson	NegBin	Poisson
(i) SAR-share moderator				
SWH × SAR share	-1.466^{***}	-0.834^{***}	-0.862^{*}	-0.980^{**}
	(0.379)	(0.209)	(0.402)	(0.306)
(ii) SAR-persons moderator				
SWH × SAR persons	-1.177^{*}	-0.687^{**}	-0.853	-1.019^{**}
	(0.534)	(0.261)	(0.492)	(0.320)
Month–year fixed effects	Yes	Yes	Yes	Yes
Newey–West SEs (lag 14)	Yes	Yes	Yes	Yes
Observations (days)	2,995	2,995	3,038	3,038

Stars denote two-sided p-values: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
Newey–West standard errors in parentheses.
SAR share is $\text{sar\_share}_{t-7:t-1}$; SAR persons is $\log(1 + \text{sar\_persons}_{t-7:t-1})$.

A cross-check estimates Equation 5 on daily Frontex SAR-event counts, independent of recorded deaths. The SWH-by-post-MoU coefficient is negative on SAR events but indistinguishable from zero on non-SAR events, locating the moderation pattern in SAR activity rather than in a general contraction of Frontex-recorded events (Appendix Table A5).

5.4 The shift is gradual, not a sharp break

The binary post-MoU specification is useful as a summary, but the regime developed over time through Libyan interceptions, NGO restrictions, and closed-port measures. Two decompositions relax the single-cut specification.

The first re-estimates Equation 5 with the SWH slope interacted with three policy regimes from Figure 2: SAR plus border control, MoU plus NGO containment, and the Lamorgese partial rollback. The model is fit without a crossing-volume control and clipped before the January 2023 Piantedosi decree so that the rollback period is not mixed with the subsequent restriction phase. Period-specific gradients are reported in Table 5.

The pre-MoU slope is consistently and significantly negative across both sources. During the MoU plus NGO-containment period, the slope turns positive in both sources, with

Table 5: Period-specific SWH–mortality gradient. NegBin with month–year FE and NW(14) SEs. UNITED is fit from 2013-01-01 to take advantage of its longer span; IOM is fit from 2014-01-01, when MMP coverage begins. Both samples end on the eve of the January 2023 Piantedosi decree.

SAR/policy regime	UNITED (2013–2023)	IOM (2014–2023)
1. SAR + border control (through Feb 2017)	-3.025^{***}	-2.108^{**}
	(0.677)	(0.693)
2. MoU + NGO containment (Feb 2017–Oct 2020)	$+1.859^{*}$	+1.202
	(0.764)	(0.628)
3. Lamorgese rollback (Oct 2020–Jan 2023)	−0.090	−0.657
	(0.570)	(0.707)
Wald H_0: all equal, $\chi^2(2)$	23.51	12.94
p-value	< 0.001	0.002
Observations (days)	3,653	3,288

Stars denote two-sided p-values: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
Newey–West standard errors in parentheses.

UNITED significant and IOM at the conventional threshold. During the Lamorgese rollback period, the estimates lie close to zero. The Wald tests reject equality of the three slopes. Within the pre-MoU regime, separating the Mare Nostrum sub-period from the subsequent Triton/Sophia phase produces gradients of nearly identical magnitude in UNITED; for IOM the Triton/Sophia phase carries the signal alone (Mare Nostrum is too thin on the Frontex-bounded sample).

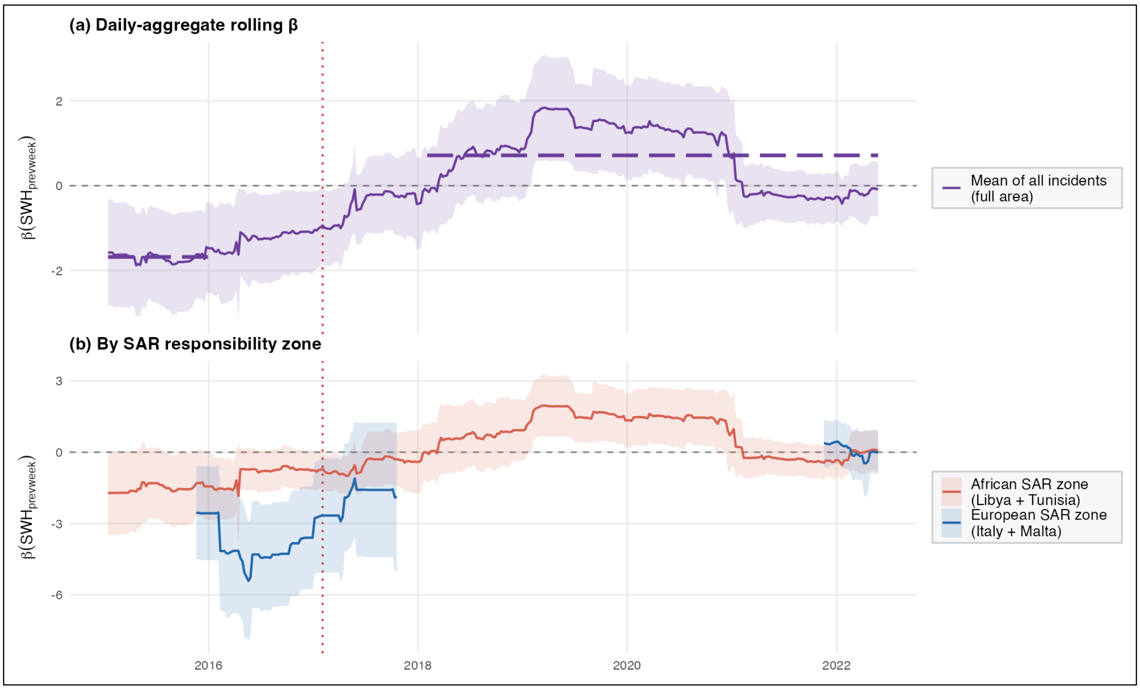

Figure 5: Rolling-window SWH–mortality gradients on UNITED deaths (730-day windows, weekly step, Poisson QMLE with month–year fixed effects and Newey–West standard errors with a 14-day bandwidth). Panel (a) plots the daily-aggregate route series across all incidents; the long-dashed horizontal segments mark the unweighted mean across windows fully pre- (left) and fully post-MoU (right). Panel (b) decomposes the rolling estimate by SAR responsibility zone: the African bloc combines Libyan and Tunisian waters, the European bloc combines Italian and Maltese waters. The red dotted vertical marks the MoU signing date (2 February 2017). Source: own elaboration.

Figure 5 confirms the same picture from a time-varying angle. Panel (a), a 730-day rolling Poisson estimate of the unconditional SWH slope on UNITED deaths across the full corridor, transitions from values around -2 pre-MoU to close to $+1$ during the MoU and NGO-containment period — with the cross-over spread across 2017–2018 rather than concentrated at the signing date — and softens back toward zero during the Lamorgese rollback.

Panel (b) decomposes the same rolling estimate by SAR responsibility zone. The European zone (Italian plus Maltese waters) shows the larger shift: a strongly negative pre-MoU gradient near -4 moves close to zero by 2019. The African zone (Libyan plus Tunisian waters), where the MoU reassigned rescue responsibility to the LCG and TCG, also rises across the regime change, from values near -1 pre-MoU to slightly positive during the MoU and NGO-containment period. Rolling-window standard errors are wider than the full-sample estimates; short-run wiggles inside the shaded bands should be read as descriptive.

6 Conclusion and Further Work

This thesis asks how the association between sea state and recorded deaths on the Central Mediterranean Route changed after the post-2017 interception regime initiated by the Italy–Libya MoU. The empirical target is the change in the daily slope linking lagged SWH to recorded mortality—not death levels or lives saved by rescue—estimated with month–year fixed effects on a 2014–May 2023 route-level panel. The mechanism analysis tests whether SAR availability moves the gradient in the direction predicted by the risk-reduction account.

The results support this interpretation in three ways. First, the main reduced-form estimates show a negative pre-MoU SWH–mortality slope and a positive post-MoU shift in both mortality sources and count-model families. Second, the shift remains after conditioning on crossing volume, boat composition, conflict pressure, and the exclusion of the Mare Nostrum sub-period. Third, the mechanism checks move as expected: higher recorded SAR activity is associated with a flatter weather–mortality gradient, and after 2017 rougher seas predict fewer SAR events but not fewer non-SAR events.

In the primary UNITED negative-binomial model, the pre-MoU slope is -2.829 (SE 0.723), the post-MoU shift is $+3.629$ (SE 0.820), and the implied post-MoU slope is $+0.800$ (SE 0.388): rougher seas predict a roughly 94% decrease in expected recorded deaths before the MoU but a more than doubling after. The UNITED Poisson estimate and the IOM comparison reproduce the same sign pattern, and conditioning on the lower-bound crossing-volume measure does not absorb the pattern (volume-controlled Poisson shifts of $+1.986$, SE 0.608, for UNITED and $+2.342$, SE 0.933, for IOM).

The mechanism and timing evidence remain reduced-form but point the same way. Weekly SAR proxies moderate the weather–mortality slope as predicted: SWH $\times$ SAR-share interactions of -1.466 and -0.834 for UNITED NegBin and Poisson, corresponding SAR-persons interactions of -1.177 and -0.687, with IOM estimates also negative. Period-specific estimates are strongly negative before the MoU, positive during MoU plus NGO containment, and back toward zero during the Lamorgese rollback. The result looks less like an instantaneous break on 2 February 2017 than a gradual regime change.

These findings should be read within the limits of the design. The analysis has no untreated comparison route, so it cannot directly estimate how the same route's weather–mortality gradient would have evolved absent the post-MoU regime. The SAR proxies are themselves endogenous to crossings, distress, detection, and operational choices, and both recorded deaths and the constructed crossing measure are incomplete relative to the full population at risk. The estimates therefore identify a policy-regime moderation parameter, not a structural causal effect of rescue capacity on true mortality.

The restrained implication is that sea-border policy should not be evaluated only through arrivals, interceptions, or aggregate deaths. The evidence is consistent with SAR acting as a safety buffer: when rescue is more available, rough seas translate less strongly into recorded mortality; when rescue is reduced or redirected, that buffer weakens. This does not imply that restoring SAR would save a specific number of lives, because the design cannot hold crossings, boats, routes, reporting, and rescue deployment fixed at the same time. It does suggest that the policy increases the risk faced by people who still depart, even when crossing volumes fall.

6.1 Further work

A first priority is better outcome measurement. Recorded mortality in UNITED, IOM, Frontex, and coast-guard registries is incomplete and partially overlapping; the true death toll exceeds what any single registry observes. Multiple systems estimation could combine imperfect lists to recover a fuller death count over space and time, allowing the slope shift to be re-estimated against a less censored outcome.

A second priority is access to more granular Frontex operational data. The PAD-194 release records detection actors, SAR status, and event types, but not vessel-level rescue tracks, response times, asset deployments, or administrative interruptions such as port-assignment delays and impoundments. These records, paired with plausibly exogenous interruptions to rescue capacity, could turn SAR availability from a weekly proxy into a measurable, time-varying treatment.

Finally, future work should develop a stronger identification strategy for the per-crossing-risk channel, moving the estimand from a regime-level moderation parameter toward a credible causal effect of rescue availability on mortality.

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Appendix

A. Code, data, and replication package

The full replication package is publicly available at https://github.com/giocopp/thesis-cmr-mortality. The repository contains the R analysis pipeline under `analysis/R/`, organised in stages (`01_download/` through `06_robustness/`) that build the daily panel, estimate the count models, and produce every figure and table cited in this thesis. Processed data files used by the regression and robustness scripts are tracked under `data/processed/`. Among the raw inputs, ACLED, IMO GISIS SAR boundaries, and UNHCR daily arrivals are tracked under `data/raw/`; ERA5 sea-state files are not tracked because of their size, and UNITED, IOM MMP, and Frontex PAD-194 are not redistributed in line with each provider's terms. These four sources are documented in `analysis/R/01_download/` and are available from the author on request. The top-level `analysis/R/run_all.R` script restores the pinned R-package environment from `renv.lock` and runs the pipeline end to end.

B. Spatial aggregation of sea state

The primary regressor is the simple spatial mean of SWH over all ocean cells inside the analysis polygon. This choice aggregates corridor-wide sea conditions rather than weather at any single incident location. The aggregation is checked in two ways: first by measuring how closely cell-level daily SWH series move with the polygon mean, and second by re-estimating the primary model after weighting SWH toward cells with more recorded incidents.

The analysis polygon contains 179 ERA5 ocean cells. Pairwise cell-to-cell correlations average $r = 0.743$ (median 0.763) across 15,931 cell pairs. Because distant cells are less synchronized, the more relevant diagnostic is each cell's correlation with the polygon-mean daily SWH: the mean is 0.863, the median is 0.882, and the 10th percentile is 0.755. The left panel of Figure A1 maps these cell-level correlations; the right panel maps the static incident-density weights used for the robustness check.

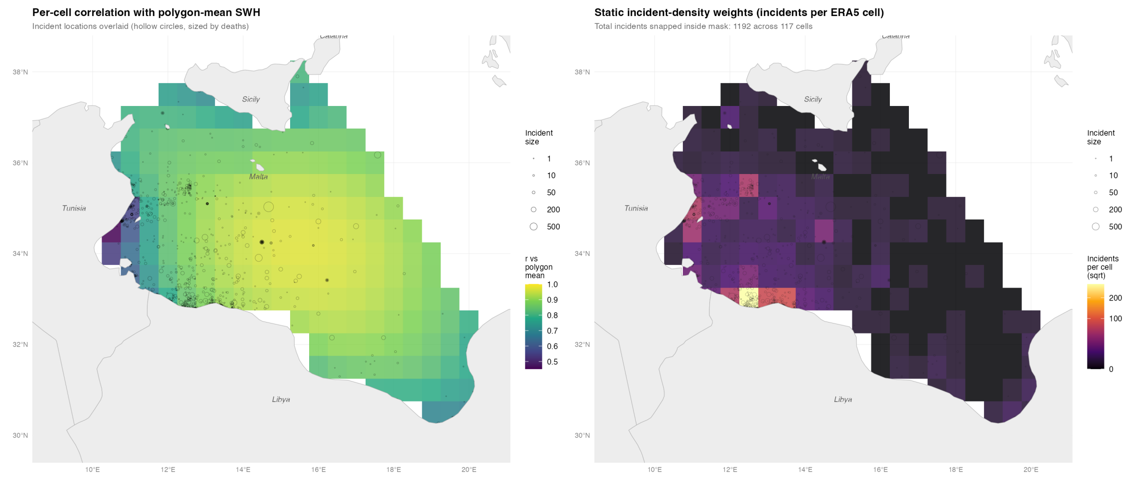

Figure A1: Spatial SWH aggregation diagnostics. The left panel maps each ERA5 cell's correlation with the polygon-mean SWH series. The right panel maps static incident-density weights, measured as IOM incidents snapped to each ERA5 cell. Source: own elaboration.

Weighted daily SWH series are constructed by assigning recorded incidents to the nearest ERA5 cell. The weights are fixed over the full sample rather than varying by day. The incident-count-weighted series has correlation 0.927 with the polygon mean; a death-count-weighted version has correlation 0.954. These alternatives concentrate weight on cells where recorded incidents cluster, but they do not condition on same-day outcomes.

Table A1 re-estimates the primary count specification with the incident-count-weighted five-day lagged SWH measure. For UNITED, the slope-shift coefficient barely changes: from +3.629 to +3.708 under negative binomial, and from +1.632 to +1.586 under Poisson. The IOM comparison estimates remain positive and statistically significant, although their magnitudes move more. The primary UNITED result is therefore not an artifact of using an unweighted polygon mean rather than a hotspot-weighted sea-state measure.

Table A1: Primary slope-shift estimates using polygon-mean and incident-count-weighted SWH.

Source	Family	Polygon mean β_3	Incident-weighted β_3	Change
UNITED	NegBin	+3.629 (0.820)	+3.708 (0.972)	+0.079
UNITED	Poisson	+1.632 (0.555)	+1.586 (0.588)	−0.046
IOM	NegBin	+2.478 (0.785)	+3.242 (0.908)	+0.764
IOM	Poisson	+2.053 (0.840)	+2.449 (0.939)	+0.396

Standard errors in parentheses. Month-year fixed effects and Newey-West SEs (lag 14). Incident weights are fixed over the full sample and do not vary by day.

C. Exposure-window sensitivity

Table A2 reports the SWH×Post-MoU shift coefficient across past and future exposure windows on a shared sample of 3,417 days (15 January 2014 to 24 May 2023). All specifications include month-year fixed effects and Newey-West (lag 14) standard errors. Past windows of 1, 3, 5, and 7 days return positive and significant shifts for UNITED in both negative binomial and Poisson families. Future-window placebos on UNITED are uniformly indistinguishable from zero. Among the IOM future placebos, the one-day lead under negative binomial is the single significant cell; the remaining seven IOM future-window coefficients are not. The choice of past window therefore does not drive the result, and the only future-window leakage is confined to IOM and to a one-day lead that overlaps with the primary five-day window through the strong day-to-day autocorrelation of SWH.

Table A2: SWH-window grid: SWH×Post-MoU shift coefficient (β_3).

Source	Window	NegBin		Poisson	
		β_3	SE	β_3	SE
Past (lagged) windows					
UNITED	lag 1d	$+2.172^{**}$	0.673	$+1.022^{*}$	0.495
UNITED	lag 1-3d	$+2.876^{***}$	0.677	$+1.554^{**}$	0.539
UNITED	lag 1-5d	$+3.663^{***}$	0.820	$+1.651^{**}$	0.555
UNITED	lag 1-7d	$+3.622^{***}$	1.085	$+1.717^{**}$	0.660
IOM	lag 1d	$+0.376$	0.626	$+1.366^{*}$	0.542
IOM	lag 1-3d	$+1.378^{*}$	0.572	$+1.770^{*}$	0.797
IOM	lag 1-5d	$+2.511^{**}$	0.783	$+2.077^{*}$	0.840
IOM	lag 1-7d	$+2.432^{*}$	1.025	$+1.997^{*}$	0.916
Future (placebo) windows					
UNITED	lead 1d	$+0.928$	0.665	$+0.687$	0.925
UNITED	lead 1-3d	-0.161	0.791	-0.140	0.655
UNITED	lead 1-5d	-0.336	0.867	-0.030	0.598
UNITED	lead 1-7d	-0.436	1.331	-0.031	0.577
IOM	lead 1d	$+2.063^{**}$	0.716	$+0.370$	0.901
IOM	lead 1-3d	$+0.951$	0.682	$+0.103$	0.517
IOM	lead 1-5d	$+0.228$	0.865	$+0.355$	0.503
IOM	lead 1-7d	-0.395	1.069	$+0.086$	0.576

SE columns report Newey–West standard errors. Stars: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Shared sample: 3,417 days.

D. Cross-source comparison

UNITED and IOM are independently collected, so their cross-source agreement is a check on whether the slope shift is a feature of either registry's recording channel. The daily and monthly distributions of route-sample deaths are tightly correlated at monthly frequency ($r = 0.91$) but only moderately correlated day by day ($r = 0.71$) over 2014–2023. The mean daily count is 5.92 for UNITED and 4.45 for IOM, with comparable right tails (95th percentiles at 31.1 and 20; daily maxima at 1,101 and 1,022).

The gap between daily and monthly alignment matters because the empirical design uses daily weather variation. IOM records fewer total deaths than UNITED in the route sample but spreads them over more nonzero death days, so its daily series is more diffuse and source-specific differences in event inclusion or event dating matter more for the estimated SWH gradient. A record-level cross-check also shows that the two registries are not duplicates: many events appear in only one source under the matching rule used here. This is why the IOM estimates are reported as a corroborating comparison rather than as the primary source.

E. Fixed-effects and inference robustness

Table A3 reports the IOM negative binomial primary specification under alternative fixed-effects and variance choices. The month-year FE specification absorbs 51.1% of SWH variance, the highest of the six FE grids tested; coarser FE deliver smaller and less stable β_3 estimates. The point estimate is invariant to the choice of Newey-West bandwidth between 7 and 21 days and to clustering by month-year or by year. Sample restrictions (dropping zero-death days or capping at 100 deaths) attenuate the magnitude but preserve the sign.

Table A3: Fixed-effects and inference robustness, IOM NegBin $\beta_3 = \text{SWH}_{t-1:t-5} \times$ Post-MoU.

Variant	β_3	SE	p
(i) FE specifications (NegBin, NW(14))			
year	+0.873	0.404	0.031
month-of-year	−0.473	0.210	0.024
year + month	+0.790	0.415	0.057
quarter-year	+1.213	0.443	0.006
month-year (primary)	+2.478	0.785	0.002
month-year + day-of-week	+2.390	0.793	0.003
(ii) SE variants on month-year FE			
Newey-West, lag 7	+2.478	0.687	0.000
Newey-West, lag 14 (primary)	+2.478	0.785	0.002
Newey-West, lag 21	+2.478	0.851	0.004
Cluster by month-year	+2.478	1.037	0.017
Cluster by year	+2.478	1.277	0.052
(iii) Sample restrictions			
Full sample, $N = 3286$	+2.478	0.785	0.002
Drop zero-death days, $N = 601$	+1.067	0.896	0.234
Cap deaths at 100, $N = 3197$	+1.933	0.577	0.001

SE column reports the standard error from the variance estimator named in each row.

F. Boat composition controls

Table A4 extends the volume-controlled Poisson specification with inflatable and wooden boat shares (Frontex-observed), plus a SWH×inflatable-share interaction in the spirit of Deiana et al. (2024). The boat-observable sample is smaller than the main panel (1,732 UNITED days, 1,729 IOM days), because boat-type coding is incomplete in Frontex JORA. The slope shift is unchanged in sign and magnitude when boat composition enters as an additive control; the SWH×inflatable-share interaction itself is not separately identified.

Table A4: Volume-controlled Poisson with boat-composition controls. Boat-observable sample: 1,732 days (UNITED), 1,729 days (IOM).

	V1: baseline	V2: + boat shares	V3: + boat × SWH
UNITED			
$\text{SWH}_{t-1:t-5}$	-1.700^{**}	-1.712^{***}	-2.316^{**}
	(0.553)	(0.478)	(0.733)
SWH × Post-MoU	$+2.402^{***}$	$+2.466^{***}$	$+2.635^{***}$
	(0.662)	(0.619)	(0.630)
$\log C_t$	$+0.535^{***}$	$+0.555^{***}$	$+0.549^{***}$
	(0.090)	(0.094)	(0.085)
SWH × inflatable share			+1.073
			(0.873)
IOM (comparison)			
$\text{SWH}_{t-1:t-5}$	-2.077^{*}	-2.091^{*}	-3.199^{*}
	(0.954)	(0.886)	(1.261)
SWH × Post-MoU	$+2.570^{*}$	$+2.643^{*}$	$+2.847^{**}$
	(1.045)	(1.038)	(0.945)
$\log C_t$	$+0.312^{**}$	$+0.331^{**}$	$+0.327^{**}$
	(0.109)	(0.121)	(0.110)
SWH × inflatable share			+1.804
			(1.058)
Month-year FE	Yes	Yes	Yes
Newey-West SEs (lag 14)	Yes	Yes	Yes

Newey–West standard errors in parentheses. Stars: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

G. Frontex SAR vs non-SAR event placebo

Table A5 reports a behavioral cross-check on the SAR channel. Re-estimating the primary specification with the daily number of Frontex SAR events as the outcome yields a strongly negative SWH×Post-MoU coefficient, while the same specification on non-SAR events as a placebo channel is indistinguishable from zero. The contrast locates the post-MoU moderation pattern in SAR activity rather than in a general contraction of Frontex-recorded events.

Table A5: SWH×Post-MoU shift on Frontex daily event counts.

	SAR events		Non-SAR (placebo)	
	NegBin	Poisson	NegBin	Poisson
$\text{SWH}_{t-1:t-5}$	+0.240	+0.302	−0.391	−0.245
	(0.183)	(0.158)	(0.206)	(0.207)
SWH × Post-MoU	-0.660^{**}	-0.854^{***}	+0.173	+0.223
	(0.235)	(0.221)	(0.254)	(0.261)
Month-year FE	Yes	Yes	Yes	Yes
Newey-West SEs (lag 14)	Yes	Yes	Yes	Yes
Observations (days)	3,407	3,407	3,438	3,438

Newey–West standard errors in parentheses. Stars: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

H. Push-factor decomposition

Table A6 reports the SWH×Post-MoU shift under a triple-interaction specification with the one-week-lagged Libya conflict composite (ACLED battles, explosions, and violence against civilians, standardized). The shift coefficient remains positive and significant across all four model-source pairs. The triple interaction itself is positive in all four pairs, though significant only for the IOM negative binomial estimate, suggesting that the slope shift is, if anything, amplified rather than absorbed by Libyan conflict pressure.

Table A6: Triple interaction SWH×Post-MoU×Libya conflict.

	UNITED		IOM	
	NegBin	Poisson	NegBin	Poisson
$\text{SWH}_{t-1:t-5}$	-2.481^{***}	-1.039^{*}	-1.647^{**}	-1.578^{**}
	(0.567)	(0.446)	(0.603)	(0.566)
SWH × Post-MoU	$+3.890^{***}$	$+1.347^{*}$	$+2.364^{**}$	$+1.629^{*}$
	(0.853)	(0.541)	(0.783)	(0.643)
SWH × Post-MoU × Libya conflict	+1.880	+1.235	$+1.951^{*}$	+1.766
	(1.031)	(0.640)	(0.992)	(0.917)
Month-year FE	Yes	Yes	Yes	Yes
Newey-West SEs (lag 14)	Yes	Yes	Yes	Yes
Observations (days)	3,197	3,197	3,286	3,286

Libya conflict is the standardized one-week lag of ACLED battles + explosions + VAC.
Newey–West standard errors in parentheses. Stars: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.